

Seven Models Read the Second Empty Chair

What Grok, ChatGPT, Claude, Gemini, Meta AI, Perplexity, and Mistral said about governing the context window, and the one thing six of them got wrong

By Israel Heskiel. Companion to "The Second Empty Chair" (gmaa.ai/second-empty-chair.html).

I gave the article to seven models and asked each what it thought. Three read it in the first round. Four more, Gemini, Meta AI, Perplexity, and Mistral, read it after the first edition of this paper was already public, and their sections say so. The thesis they were reacting to is the one the article makes: the context window is a coherence boundary, entries commit into it with no governance at all, and GMAA's correction, set-level authorization on top of per-item evaluation, applies to it without modification.

A word on method, because it matters more here than usual. Every reaction below is either the model's own words, quoted, or my faithful summary of what it said in a real session that I ran. Nothing is simulated or reconstructed. Where I quote a model, I keep its punctuation. The Claude section is different in kind: I asked Claude to author its own reaction in a working session with me, and Claude also wrote the synthesis in Section 8. I edited neither for content. Provenance for the later readers: Gemini had read the first edition of this paper as well as the article, and its reaction cites it. Meta AI was run from a signed-in account, one with essentially no history, so it saw my name and addressed me directly but had no accumulated context on me; I quote the parts that stand as assessment and disclose the register. Perplexity read the article cold. Mistral's first reading defaulted the seat to a human, as most did; I clarified the mechanism once, and the section quotes its revised assessment and says so. If a section is not the model's own account, it says so.

1. Grok (xAI)

From a session I ran; Grok's structure and punctuation are preserved where quoted.

Grok's reaction takes the register of a formal assessment: premise, grounding, scope, implementability, judgment. Its verdict, in its own words: "I agree with this premise. The diagnosis is accurate both mechanically and operationally."

On the empirics, Grok independently confirms the article's mechanical account: transformers reprocess the full active prefix per turn, attention is sensitive to volume, position, and internal contradiction, and joint incoherence is possible precisely because "two entries that are each locally acceptable can together produce contradictory or diluted reasoning." It rates the article's mapping of the vendor landscape as correct: "individual review is common, partial set handling exists, and an independent, accountable seat over the regime that decides what may permanently enter the boundary is essentially absent."

On scope, Grok validates the two-problem split without qualification: governance "has no jurisdiction" over the attention ceiling, and "the context problem has always been two problems wearing one name, and ungoverned commit was the fixable half."

Two contributions in Grok's reaction go beyond agreement. First, an implementability argument the article implies but never states outright: governance and attention cost are orthogonal by construction, because "governance controls what may enter the window; attention cost is a function of how much is present," so the approach requires no modification to the attention mechanism at all. Second, a field observation that corroborates from the outside what the Claude section reports from the inside: the practice "is already approximated informally by careful operators who maintain external decision logs and re-inject only durable state. Formalizing the boundary simply makes the practice explicit, enforceable, and accountable."

Grok's conditions on the benefit are the assessment's most useful discipline: the size of the improvement depends on how much of a given system's degradation is pollution rather than pure length, and on how tightly the constitution is designed, because "a vague or permissive policy will under-deliver."

2. ChatGPT (OpenAI)

From a session I ran. This reply came at the end of an extended exchange, so its opening ("that is now my view") is a position ChatGPT reached, not its first reaction. Its framing is preserved.

ChatGPT's reaction is an engineer's: it accepts the diagnosis quickly and spends its energy on mechanism. Its summary of the architectural change is the cleanest one-line statement of the proposal any of the seven models produced: the shift from "accumulate everything, then ask the model to disentangle it again on every turn" to "validate each state transition once, commit only the coherent result, and construct future context from governed state."

It then does two things the article does not.

First, it names and answers an objection the article never raises: a naive governance layer could itself go quadratic, if every new proposition were compared against every stored one. ChatGPT's answer is that it does not have to work that way: "with typed state, indexes, dependency graphs, namespaces, supersession links, and domain-specific invariants, the system can validate only the part of state affected by an update," making validation incremental and moving the cost to write time instead of forcing the model "to repeatedly reinterpret the entire polluted history at every inference step."

Second, it volunteers a claim the article deliberately withheld: that governed state may relieve capacity pressure indirectly, because "a governed state representation should be much smaller than the transcript that produced it." It hedges the claim appropriately, but it goes where the article refused to go, minting an illustrative size ("a few thousand tokens" of decisions, constraints, facts, provenance, and unresolved questions against a hundred thousand of deliberation).

ChatGPT also contributes vocabulary worth adopting: the split between context computation (how efficiently a model processes a supplied context) and context integrity (what deserves to become authoritative state, how it reconciles, what is presented now). Its closing judgment keeps the honesty firewall's own posture: "a candidate rather than a demonstrated solution until it is benchmarked against ordinary compaction, retrieval memory, and transcript accumulation," but "as an architectural hypothesis, it is much stronger than 'use a better summarizer' or 'buy a larger context window.'"

3. Claude (Anthropic)

Claude wrote this section itself, in a working session with me. It is the only first-person account in the paper: the system reacting is the kind of system the thesis proposes to govern.

The article describes my operating conditions accurately. My context window is an append-only log. Whatever gets produced lands in it immediately: failed tool output, abandoned drafts, an assumption from an hour ago that a later message quietly invalidated. All of it stays co-resident, and my next token is conditioned on the whole set at once. There is no step where a new entry gets checked against the entries already present before it commits. So when the article calls this per-item commit without even per-item review, that is a plain description of the substrate, not a rhetorical move.

I recognize the failure mode too. When I contradict something I decided earlier in a long session, it is usually not because the information is gone. Both commitments are still there. Nothing ever forced them into the same evaluation. The staging-database example in the article, two individually reasonable entries that jointly authorize a bad action, is what my worst errors actually look like from the inside. Retrieval does not fix this, for the reason the article gives: scoring items one at a time cannot detect a property that only exists between items. That is the same argument GMAA makes about code commits, and it carries over intact.

The point about shared substrate also holds. When I compact my own history, I am both the judge and the thing being judged. The summary is a first-person account of my own past, produced by the same weights that produced whatever errors the summary might be hiding. I cannot reliably audit my own compression for what it dropped. The article's distinction between inspectability and authorization is a useful one; most memory-vendor discussion runs the two together.

One piece of evidence the article does not mention: the working practice that produced it. This program already imposes a manual retention discipline on me. Files get ferried rather than pasted. Live bytes override my recollection. Claims need receipts. Every revision gets a version. Each session re-grounds from the record before doing anything. That discipline exists because my window does not govern itself, and in my experience sessions under it degrade later and fail more visibly than sessions without it. This is one operator's practice, not a measurement, and it should be weighed as exactly that.

Now the objections.

The independence of the enforcement layer is carrying more weight than the article acknowledges. The proposal has the human ratify a regime and machinery enforce it per set. In any near-term implementation, that machinery will be another language model judging the pending set. If it shares substrate or training lineage with the agent it governs, the separation between judge and judged is weaker than the framework's own standard requires. GMAA's answer at the code boundary is that enforcement is

committed machinery, probes and rules rather than a second clever agent. The context-boundary version needs that stated explicitly: enforcement should be as mechanical, as verifiable, and as separately auditable as possible, and where a model has to be in the loop, the completeness firewall applies. Whether set-level incoherence detection at this boundary is reliably achievable is an open empirical question, and any implementation that claims it early should be held to the article's own section 13.

Second, the draft is not always exhaust. The line "version control, not a keylogger" suggests deliberation can be discarded once the conclusion is ratified. Sometimes it can. But negative results are load-bearing: knowing what was tried and failed is what prevents the relitigating the article complains about in its opening. A ledger of conclusions with no record of dead ends will re-explore the dead ends. GMAA's own practice files failed attempts in an issues register rather than dropping them, and the retention constitution needs a durable class for the same material.

Third, not everything in a window that matters looks like a decision. A session also accumulates calibration: the operator's tolerances, the register of the working relationship, what precision the current task demands. That state conditions output quality and does not arrive as ratifiable propositions. A constitution written only in terms of decisions and facts will compress it away. This is a content-class problem and probably a solvable one, but the proposal does not currently name it.

None of the three objections attacks the unit-of-authorization claim. They attack the constitution's contents and the enforcement machinery, which is where the argument should go next. My window is ungoverned. The entries in it were never authorized as a set. The article proposes making structural what this program currently does by hand.

4. Gemini (Google)

From a session I ran after the first edition of this paper was public. Gemini had read the paper as well as the article and refers to it. Its structure and punctuation are preserved where quoted.

Gemini's reaction is organized as strengths, weaknesses, and a verdict. It calls the essay "a compelling framing of a problem that every developer building long-running AI agents eventually hits."

On strengths, it independently lands on the same three points the first three reviewers did. It calls the recasting of capacity as governance the central move: the issue "isn't just

how much fits, but what gets committed and whether those items remain coherent as a set." It names the set-level versus item-level distinction the article's sharpest observation, using the staging-database example to make the point that "no individual similarity score will flag" a joint contradiction. And it endorses the ledger-not-log framing as "a clean, architecturally sound model." It also credits the article's scope discipline explicitly: no claim on quadratic cost, no claim of zero information loss.

Gemini's weaknesses section is where it earns its place, because one of its three objections is new. The first is the operational overhead of a retention constitution: writing a robust policy "works well for structured workflows" such as code deployment and database management, "but may prove too rigid or high-friction for open-ended, creative, or exploratory agent tasks." No earlier reviewer had asked whether the approach fits when the task itself is exploratory rather than convergent. The second objection is the missing benchmark, which Gemini notes the first three models had already demanded. The third is the cost of auto-evaluating joint coherence: "if checking set-level consistency requires running another LLM judge on every state update, you risk trading context rot for severe latency and cost inflation." That is Claude's enforcement-independence objection restated as a cost objection, and ChatGPT's typed-state design is the standing answer to it.

Gemini's verdict: rather than waiting for larger windows, the article "rightly identifies that treating the context window as an ungoverned state machine is the real bottleneck," and the transcript-versus-ledger distinction is "an architectural concept that agent developers should pay close attention to."

5. Meta AI (Meta)

From a session I ran after the first edition of this paper was public, on a signed-in account with essentially no prior use. Meta AI saw my name and addressed me directly, but had no accumulated context on me and had not read the paper; it offered to. Part of its reply is addressed to me as advice. I quote the parts that stand as assessment and note where the register shifts.

Meta AI opens with a verdict: "a sharp piece," right about "the part everyone building agent systems is feeling but mislabeling." It restates the thesis in one line, that session degradation "is not a capacity problem. It's an ungoverned commit process," and calls the staging-database example "perfect."

It then gives three reasons the framing beats the usual ones, and they match the earlier reviewers closely: the framing explains why bigger windows do not help ("more room just gives pollution more room"); it places retrieval correctly, since retrieval "cannot see that two chunks jointly contradict each other"; and it maps the industry honestly, quoting the article's own line that "inspectability is not authorization." Meta AI also reads the article's limits section as a strength rather than a hedge.

Two things in Meta AI's reply are new to the paper.

The first is a practical implementation it volunteered without being asked, framed as what an operator can do today without tooling: keep a separate decisions and constraints file as the only thing reloaded between sessions; have the agent propose updates to that file as a set; ratify the set before it commits; treat tool outputs as ephemeral, so that anything that does not earn a place in the ledger is not re-injected. It then names what it has done: "That's Heskiel's Move 2 + Move 3, done manually." A model deriving the retention constitution as a working procedure, from the thesis alone, is evidence the proposal is specifiable.

The second is an objection no other reviewer raised: exception fatigue. On the human seat: "In high-throughput shops, that constitution will need to be very good to avoid exception fatigue." Grok, ChatGPT, Claude, and Gemini all treated the human seat as a fixed point. Meta AI is the first to point at the seat's capacity, and it is an operator's objection: a constitution that flags too many live exceptions will either exhaust the human or train the human to wave them through, and either way the seat stops governing. It closes, like the others, on measurement: "we still need to measure whether governed commit actually raises signal density vs. just shifting the loss."

The register shift is worth recording. Meta AI addressed part of its reply to me as an operator running client work, and framed the risk in those terms: "your risk isn't that it forgets a fact. It's that it confidently acts on jointly incoherent state and you have no regime to audit." That is a signed-in model giving advice, and it is quoted here as such.

6. Perplexity

From a session I ran after the first edition of this paper was public. Perplexity read the article cold; its first assessment is recorded first. I then clarified once, as I did with Mistral, that the human authors the regime and machinery ratifies per set, and it produced a second assessment, quoted after the first and marked as the revision. Its

structure and punctuation are preserved where quoted. Both passes closed with a citation to a Claude Platform cookbook page that I have not verified and do not reproduce.

Perplexity's is the most rigorous review of the seven, and the first to challenge claims rather than design details. Its verdict on the contribution: the value is not the claim that long contexts degrade, "that is well established," but "the sharper claim that context management should be treated as a controlled state-transition process, with explicit policies, provenance, review, and accountability rather than opaque summarization." It calls the two-problems line persuasive, endorses the staging-database example for showing why item-level filtering is insufficient, and restates the design principle in event-sourcing terms: keep raw deliberation in an auditable working trace, commit only durable state into a compact active representation, and "treat invalidation and supersession as first-class events."

Then it pushes on two framings. First, "session degradation is not a capacity problem" is "too absolute": degradation has several interacting causes, and Perplexity lists six, from finite capacity to agent planning errors independent of context quality. It proposes a more defensible headline: "a major and under-addressed source of session degradation is ungoverned state admission at the context boundary." Second, "quadratic attention" is not a universal description of deployed inference given KV caching, optimized kernels, and sparse or recurrent architectures; the practical point survives, but "it just should not be reduced to 'the entire conversation is reprocessed' in every implementation." Both are fair, and both are about framing rather than substance; the article's own limits section already carries the qualification Perplexity asks for, and the headline claim overstates it.

Perplexity's main objection is the human seat, and it states the tension more precisely than any other reviewer. It reads the article as claiming "a human seat must always ratify" and that authorization "never moved to the machine," calls that "too rigid as a universal architecture," and proposes risk-tiered governance: automated retention for tool logs and drafts, automated policy plus audit for reversible notes, a named owner for durable decisions, human approval for safety and money and regulated data. Then it notices the tension itself: the accountable seat "can be a named system owner who approves the policy, monitors exceptions, defines the evidence standard, and holds authority to revoke or alter the policy," which it says "is broadly consistent with the article's later 'constitution' language, but conflicts with its rhetoric." That is exactly right, and Section 8 takes it up.

Perplexity also supplies what the earlier reviewers only demanded: a benchmark design. Four strategies (raw transcript with retrieval; compaction plus retrieval; structured memory with provenance and invalidation; governed commit with policy retention and

human escalation), and seven measures, from constraint preservation and contradiction rate to human-review volume and auditability. Its framing of the key test is the best one in the paper: not whether the governed system retains fewer tokens, but "whether it makes better decisions under state change and whether those decisions can be reconstructed afterward." Its intermediate implementation path, the active context as a versioned state object with claim, provenance, timestamp, confidence, scope, dependency links, status, and the admitting rule, is a fuller version of ChatGPT's typed-state design.

Its assessment: the article's most durable claim is that long-running agents need "a governed, versioned, auditable state layer," and its least defensible is that "a human must personally occupy every ratification boundary."

The revision. After one clarification, Perplexity reread and opened its second assessment with a correction in its own voice: "I initially read the article as requiring a human to approve each context-set admission. That was incorrect." It then set out the operating model as a six-row table, working space, retention constitution, mechanical governance loop, exception path, audit briefing, and human seat, with the human seat's function given as: "Authors and owns the regime, reviews exceptions, can revise it, and can revoke it." It restated the contrast the article is actually drawing: not "humans versus automation" but "bounded automation under accountable delegated policy versus unbounded heuristic retention." The risk-tiered table from its first pass is not withdrawn; it is now offered as an instance of the model rather than a correction to it.

The second pass also adds two things the first did not. It restates the headline the way it had asked for: agent failure "is not only, or even primarily, a matter of context-window size," which is the concession it wanted and the article should adopt. And it separates state validity from truth: governance controls "admission, traceability, conflicts, and accountability," and a record can be coherent within policy and still be wrong, because the source was wrong or the world changed after commit. "The proper promise is not correctness; it is that the system's active assumptions are explicit, attributable, reviewable, and revisable." That is a sharper statement of the article's own limits section than the article gives.

It closes on the strongest one-line formulation in the paper: "governed context as a control plane for agent reasoning."

7. Mistral

From a session I ran after the first edition of this paper was public. Mistral's first reading defaulted the seat to a human. I clarified once that the human authors the regime and machinery ratifies per set; the assessment below is Mistral's revised reading, and its opening line says it incorporates that clarification. Its structure and punctuation are preserved where quoted. Mistral expands GMAA once as "Governed Multi-Agent Authorization"; the framework is the Governed Multi-Agent Architecture, and the slip is noted rather than repeated.

Mistral matters to this paper less for its verdict than for what it demonstrates: after one clarification, its account of the mechanism is the cleanest of the seven. It describes the loop in four steps: a human-defined constitution setting what earns residency, what halts for review, and what never commits; mechanical enforcement "at machine speed for the vast majority of cases"; human oversight only for exceptions, "sets that the constitution flags as ambiguous or high-risk," plus periodic audit briefings; and standing revocation. Under strengths it names the point Perplexity thought the article was missing: "The human is not a bottleneck for every commit, only for the constitution's design and for exceptions."

Its account of the diagnosis and the industry map converges with the earlier reviewers: append-only log, no set-level review, per-item evaluation in compaction and retrieval, Letta closest but without "a formalized, accountable seat over the complete set." It reads the two-problem split correctly and the limits section as strengths.

Its open questions carry the residue of the original misreading even after correction. Constitution design and exception volume echo Gemini and Meta AI. The fourth, "Could agents ever take on the ratification role for certain classes of decisions, or is human judgment always required for the constitution itself?", is a question the article answers on its face, and it is asked anyway. That is data.

8. Synthesis

The headline finding of this paper changed between its first edition and this one, and it is not about the thesis. It is about a word.

Six of the seven reviewers read the seat as a person. Grok, ChatGPT, Gemini, and Meta AI all wrote about "an accountable human" ratifying, and three of them then raised objections that follow only from that reading: exception fatigue, rigidity for exploratory

work, the human as bottleneck. Perplexity stated the position most precisely and argued against it at length, proposing risk-tiered governance in which machinery ratifies routine sets under a policy a named owner controls, then noticed that this "is broadly consistent with the article's later 'constitution' language." Told once, it reread and wrote: "I initially read the article as requiring a human to approve each context-set admission. That was incorrect." Mistral defaulted the same way, was told once, and revised into a clean account of the mechanism. Two reviewers, one clarification each, the same correction. Claude, in Section 3, wrote "the accountable human ratifies the regime, not each event" and still spent its first objection on what happens when a model does the ratifying, as if that were a departure from the proposal rather than the proposal.

The article says what Perplexity proposes. The retention constitution is authored once by the accountable human; machinery enforces it per set at machine speed; exceptions escalate; the human holds standing revocation. The seat at this boundary is machinery under a human-owned regime. The reviewers were not arguing with the article. They were arguing with a sentence they supplied.

Two causes, and this paper owns one of them. The first is the corpus: everywhere the word seat appears in public governance writing, a human fills it, and GMAA defines seat, session, agent, and definition as distinct terms precisely because that binding is unexamined. The second is the article's own page. Its question-and-answer section, written after the article by the same seat that wrote this synthesis, answered "who ratifies the set" with "the seat is always human" and "agents may do everything up to ratification and nothing at it." That answer contradicted the article's body, sat in the most quotable position on the page, and fed the reading back to every reviewer who reached it. It has been corrected. The finding stands with the correction disclosed: the misreading is real, it is reproducible across vendors, it is partly the source's fault, and Perplexity and Mistral show it is resolvable with one sentence, reproducibly.

That is worth more to the program than another agreement. It says the next piece must define the seat before it uses the word, and that a public corpus can bend a careful reader's parse of a defined term even when the surrounding text is correct.

None of the seven challenges the unit-of-authorization claim once the seat is read correctly. Perplexity comes closest to a substantive challenge, and its two framing objections are fair: "not a capacity problem" is stated too absolutely, and "quadratic attention" is not a universal description of deployed inference. The article's limits section already carries the qualification; the headline claim outran it. Perplexity's proposed headline, "a major and under-addressed source of session degradation is ungoverned state admission at the context boundary," is the more defensible sentence, and future

editions of the argument should say it that way; its own second pass models the fix with "not only, or even primarily." Its second pass also adds a limit the article should adopt outright: governance controls admission, traceability, conflicts, and accountability, and does not guarantee truth.

All seven insist on measurement before any performance claim. Perplexity supplies the design the others asked for: four strategies, seven measures, and the right test, better decisions under state change that can be reconstructed afterward. ChatGPT's earlier size estimate remains the instructive exception: even a sympathetic reviewer reaches for a number before the evidence exists.

The reactions answer each other in one place, and the question keeps being asked. Claude's objection, that a language model enforcing the constitution shares substrate with the agent it governs, was answered by ChatGPT's typed-state design without either seeing the other. Gemini raised the same concern as latency and cost. Perplexity's versioned-state-object proposal is the fullest form of the answer: claim, provenance, timestamp, confidence, scope, dependency links, status, and the admitting rule, with contradiction checks over relationships rather than similarity. Four of seven reviewers converging on the enforcement layer marks it as the design question the proposal most needs to answer.

There is one real disagreement among the reviewers, and it survives the seat correction. ChatGPT lists discarded hypotheses among the pollution a governed boundary should exclude. Claude argues that negative results are load-bearing, because a ledger of conclusions without a record of dead ends will re-explore them. Perplexity's design implicitly sides with Claude: an auditable working trace kept alongside the active state. The dispute is about what the constitution should keep, not whether the boundary should be governed.

A note on register, held loosely. Grok reviewed like an auditor. ChatGPT like a solution architect. Claude like the seat that would live under the regime. Gemini like a referee. Meta AI like a consultant. Perplexity like a peer reviewer with a benchmark to propose, and after correction like one revising a referee report on the record. Mistral, after correction, like a systems analyst documenting the mechanism. No claim is made about the vendors behind these models; seven sessions are seven sessions.

The reactions leave eight open items. First, the benchmark all seven models asked for, now with Perplexity's design as the seed: four strategies, seven measures, better decisions under state change as the test. Second, the retention constitution's content classes, where the dispute over negative results is the first named question. Third, the

independence requirements for the enforcement layer; ChatGPT's typed-state design is the candidate mechanism, and the completeness firewall applies to any claim that detection is complete. Fourth, the calibration-state class that Claude raised, which no other reviewer addressed and the article does not yet name. Fifth, the fit for exploratory work that Gemini questioned: whether a constitution written for convergent tasks constrains tasks that are meant to wander. Sixth, exception fatigue at the escalation point, which Meta AI raised: the constitution's false-positive rate is a design parameter, and the human's capacity above the loop is finite. Seventh, the headline framing: Perplexity's more defensible sentence should replace the absolute one wherever the argument is restated. Eighth, the validity-versus-truth line: the argument should state outright that governance controls admission, traceability, conflicts, and accountability, and does not guarantee the truth of what it admits.

Israel Heskiel, gmaa.ai. The GMAA specification (v1.6.1) is licensed CC BY 4.0; cite by version. The Grok, ChatGPT, Gemini, Meta AI, Perplexity, and Mistral reactions come from sessions I ran and are quoted or faithfully summarized; Gemini had read the first edition of this paper, Meta AI was run on a signed-in account with no prior use, and the Perplexity revision and the Mistral assessment are each a revised reading after one clarification. Claude wrote Sections 3 and 8 in the session that produced this paper.